Supplementary Material: Auditing SHAP Explanations Against Known Failure Mechanisms in Predictive Maintenance

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Numerical tables are generated from stored result files by the accompanying scripts. The immutable computational snapshot is `tuai-computation-5b97414ab7ef.zip`, with full SHA-256 recorded in its accompanying checksum file and per-file hashes in `SNAPSHOT_MANIFEST.json`. Source repository: `https://github.com/kowshik24/glowing-broccoli`. This local snapshot does not establish public release availability; stage numbers refer to `experiments/NN_*.py`. Holm-adjusted p-values throughout.

S1 Implementation

Classifiers Table S1 lists the settings of the seven benchmark classifiers and Table S2 their failure-class test performance (scikit-learn 1.9.1, XGBoost 3.4.1, PyTorch 2.14.0; seed 42). Precision and recall intervals are Clopper–Pearson, the others 5,000 stratified bootstrap resamples. Five-fold cross-validation of tuned XGBoost at threshold 0.5 gives failure-class F1 0.822 ± 0.013; its Brier score is 0.0090 (0.033 for a base-rate predictor) and its expected calibration error 0.008.

Table S1: Classifier settings.

Model	Settings
Logistic regression	max. 1000 iterations
SVM	radial basis function (RBF) kernel, probability outputs
k-NN	$k = 7$
Decision tree	max. depth 10
Random forest	300 trees, max. depth 12
XGBoost	300 trees, max. depth 6, learning rate 0.1
XGBoost (tuned)	200 trees, max. depth 8, learning rate 0.1, subsample 0.8, column sample 0.8, min. child weight 1, $\gamma = 0$
MLP	64 and 32 hidden units, ReLU, batch norm, dropout 0.3/0.2; Adam (learning rate 10^{-3}, weight decay 10^{-5}), binary cross-entropy, batch 64, 40 epochs

Table S2: Failure-class test performance, sorted by F1; parentheses: 95% bootstrap interval for F1.

Model	Prec.	Rec.	F1 (95% CI)	ROC-AUC
XGBoost (tuned)	**1.00**	0.79	**0.89** (0.79–0.95)	0.988
XGBoost	0.86	0.88	0.87 (0.78–0.94)	0.988
Random forest	0.79	0.88	0.83 (0.74–0.92)	**0.989**
MLP	0.65	**0.91**	0.76 (0.67–0.84)	0.988
Decision tree	0.68	0.82	0.75 (0.64–0.84)	0.927
SVM	0.42	0.88	0.57 (0.48–0.65)	0.976
k-NN	0.43	0.79	0.56 (0.46–0.65)	0.877
Logistic regression	0.24	0.85	0.37 (0.31–0.43)	0.943

Explainers and variants Interventional TreeSHAP draws 100, 500, or 1,000 background observations per fold from the original (not oversampled) training data with three background seeds, or, for TWF, 100 observations only from non-failures, only from failures, or from the oversampled training set. The pairwise reference is the nearest normal-operation training observation of each explained observation. The MLP (64 and 32 hidden units) is explained on its logit by KernelSHAP (25 background observations, 1,000 coalitions; exact with 100 or 500 background observations and all 4,096 coalitions; or 25 weighted k-means centres), Integrated Gradients (1,024 steps from the mean of 100 training observations, completeness within 5%), DeepSHAP (100 background observations), and SmoothGrad (50 samples, noise s.d. 0.15 in standardized units, times the input's difference from the same mean). LIME explains the failure probability of the XGBoost pipeline with 1,000 perturbations per observation and the fold's training data. Grouped TreeSHAP sums signed path-dependent SHAP over four groups (wear: tool wear, overstrain, its ratio; thermal: both temperatures, their difference; load: torque, speed, power; type), and the conditional group Shapley values evaluate all 16 coalitions with Gaussian kernel weights over 2,000 training observations of the fold (bandwidth 0.3 in standardized units). Every variant uses a null of the same pipeline.

S2 Primary audit pseudocode

The following algorithm describes the five-fold AI4I audit with fixed hyperparameters. CWRU instead uses four recording-group folds and recording-label permutations; TEP uses five run-group folds and within-run label permutations, as specified in the paper. Model selection is not nested in the AI4I audit.

Algorithm 1 Mechanism-level attribution audit

Require: data (x_i, y_i, z_i), expected sets E_m, pipeline $\mathcal{P}$, explainer ϕ, K, α

$A \leftarrow$ held-out $|\phi|$ from five-fold cross-fitting of $\mathcal{P}$ on (x, y)
compute R_m, D_m, C_m from A for every mechanism m (Eqs. (2)–(4) of the paper)
for $k = 1$ **to** K **do**
    $y^{(k)} \leftarrow$ random permutation of y; keep x and z
    $A^{(k)} \leftarrow$ held-out $|\phi|$ from cross-fitting $\mathcal{P}$ on $(x, y^{(k)})$
    compute $R_m^{(k)}$, $D_m^{(k)}$, $C_m^{(k)}$ from $A^{(k)}$
end for
p-values by Eq. (5) of the paper; Holm correction over mechanisms for each statistic
for each mechanism m **do**
    if $p_{C,m} < \alpha$ **then**
        flagged; report the feature with the largest excess
    else if $p_{R,m} < \alpha$ and $p_{D,m} < \alpha$ **then**
        aligned
    else
        not aligned
    end if
end for

Ensure: verdicts, flagged features and excesses, effect sizes z, bootstrap intervals

S3 Reporting checklist

The following steps summarize the audit configuration we recommend fixing and reporting.

1. **Mechanism map.** Fix E_m from rules or physics before the audit. State which inputs are engineered from a documented rule, and repeat the audit on the original inputs to show how much the engineered features contribute.

2. **Attribution.** State the explainer, the explained output (logit or probability), and the reference distribution. For questions about abnormal operation, use a normal-operation reference; when it differs from the training data, report both.

3. **Held-out explanations.** Explain each observation with a model that did not train on it; hold out whole recordings or runs when observations within them are dependent.

4. **Null.** Refit the full pipeline (preprocessing, resampling, training, explanation) on shuffled labels. Permute within groups, and shift labels in time when they are temporally dependent. Use $K \geq 200$, and $K \geq 500$ when an adjusted p lies near α.

5. **Verdicts.** Apply Holm correction across mechanisms for each statistic and report R_m, D_m, and C_m with their null means, z, and bootstrap intervals, not only the verdict.

6. **Flags.** Report the flagged feature and its excess. Treat excesses below about 0.05 as exploratory and inspect each flag for a missing trigger, a downstream effect, or a shortcut.

7. **Stability.** Repeat the audit over seeds, background samples, and at least one alternative explainer or reference, and report verdicts that change.

S4 Full audit p-values

Table S3 gives the Holm-adjusted p-values for all variants, including the selected variants in Table III of the paper. The main table omits background-size, neighbour-count, and other implementation variants to keep the main comparisons readable.

Table S3: Holm-adjusted p-values of the R, D, and C tests for every audit variant. $^{\times}$: non-significant alignment test; †: significant outside-set test; minimum attainable adjusted p: $4/(K+1)$.

		TWF			HDF			PWF			OSF		
Variant	K	R	D	C	R	D	C	R	D	C	R	D	C
Primary, seed 42	1000	$0.201^{\times}$	$0.077^{\times}$	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
single-mechanism rows	1000	$0.187^{\times}$	$0.076^{\times}$	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
null re-tuned per permutation	500	$0.447^{\times}$	$0.359^{\times}$	1.000	0.018	0.008	1.000	0.008	0.008	1.000	0.032	0.008	1.000
positive part of SHAP	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Seeds 1–4 (largest p)	200	$0.269^{\times}$	$0.129^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Raw features only	1000	$0.345^{\times}$	$0.616^{\times}$	1.000	$0.190^{\times}$	0.004	1.000	0.004	0.004	1.000	0.004	0.048	1.000
In-sample SHAP	1000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
Unflagged failures removed	200	$0.179^{\times}$	$0.075^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Label = OR of flags	200	$0.308^{\times}$	$0.144^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Class weights, no oversampling	200	$0.085^{\times}$	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Focal loss, no oversampling	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Borderline-SMOTE, 3 neighbours	200	$0.184^{\times}$	0.0498	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Borderline-SMOTE, 10 neighbours	200	$0.269^{\times}$	$0.149^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
SMOTE	200	$0.169^{\times}$	$0.085^{\times}$	1.000	$0.109^{\times}$	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
ADASYN	200	$0.159^{\times}$	$0.090^{\times}$	0.955	$0.109^{\times}$	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
E_m without derived features	200	$0.468^{\times}$	$0.209^{\times}$	0.478	0.040	0.020	$0.020^{\dagger}$	$0.179^{\times}$	$1.000^{\times}$	$0.020^{\dagger}$	$0.776^{\times}$	$1.000^{\times}$	$0.020^{\dagger}$
E_m without top trigger	200	$0.701^{\times}$	$0.209^{\times}$	0.478	$0.995^{\times}$	0.020	$0.020^{\dagger}$	$0.239^{\times}$	$0.935^{\times}$	$0.020^{\dagger}$	$0.995^{\times}$	$0.657^{\times}$	$0.020^{\dagger}$
Interventional, background 100	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
failure-only background	200	$0.995^{\times}$	$0.945^{\times}$	$0.020^{\dagger}$	$0.826^{\times}$	$0.149^{\times}$	1.000	0.045	$0.090^{\times}$	1.000	0.040	0.020	1.000
non-failure background	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
oversampled background	200	0.045	0.020	0.896	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Pairwise (nearest normal reference)	200	0.020	0.020	0.498	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Random forest	200	$0.617^{\times}$	$0.398^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.040	0.020	1.000
Grouped TreeSHAP (4 groups)	200	$0.816^{\times}$	$1.000^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.030	1.000
Conditional group Shapley	200	0.020	0.020	1.000	0.020	0.040	0.517	0.020	0.020	1.000	0.020	0.040	1.000
LIME	200	0.040	0.020	1.000	0.020	0.020	1.000	0.020	$0.159^{\times}$	1.000	0.040	0.020	0.597
MLP, KernelSHAP	200	$0.975^{\times}$	$0.831^{\times}$	0.975	$0.388^{\times}$	$0.189^{\times}$	1.000	$0.358^{\times}$	0.020	1.000	$0.358^{\times}$	$0.075^{\times}$	1.000
exact, background 100	200	$1.000^{\times}$	$0.915^{\times}$	0.756	$0.557^{\times}$	$0.219^{\times}$	0.965	$0.557^{\times}$	0.020	0.965	$0.557^{\times}$	$0.149^{\times}$	0.851
exact, background 500	100	$0.990^{\times}$	$0.881^{\times}$	0.792	$0.683^{\times}$	$0.337^{\times}$	1.000	$0.683^{\times}$	0.040	1.000	$0.277^{\times}$	$0.089^{\times}$	1.000
k-means background	200	$0.980^{\times}$	$0.900^{\times}$	0.637	$0.547^{\times}$	$0.299^{\times}$	1.000	$0.493^{\times}$	0.045	1.000	$0.080^{\times}$	0.020	1.000
MLP, Integrated Gradients	200	$0.338^{\times}$	$0.100^{\times}$	0.522	0.020	0.020	1.000	0.0498	$0.100^{\times}$	0.478	0.030	0.020	1.000
MLP, DeepSHAP	200	$0.920^{\times}$	$0.368^{\times}$	0.478	0.020	0.020	1.000	$0.104^{\times}$	0.045	1.000	$0.129^{\times}$	0.045	1.000
MLP, SmoothGrad	200	$0.970^{\times}$	$0.677^{\times}$	$0.040^{\dagger}$	0.020	0.020	1.000	0.0498	0.020	1.000	0.020	0.020	1.000
Permuted labels (sanity)	200	$1.000^{\times}$	$0.731^{\times}$	1.000	$0.199^{\times}$	$0.557^{\times}$	1.000	$1.000^{\times}$	$0.731^{\times}$	1.000	$1.000^{\times}$	$0.731^{\times}$	0.637

S5 Choice of K

The Monte Carlo standard error of a raw p-value near $\alpha = 0.05$ is $\sqrt{\alpha(1-\alpha)/K}$: 0.022, 0.015, 0.010, and 0.007 for $K = 100$, 200, 500, and 1,000. This is the error of a raw p-value; the uncertainty of a Holm-adjusted decision also depends on the other mechanisms' p-values. Primary verdicts are identical at $K = 100$, 200, 500, and 1,000, so we recommend $K \geq 200$, and $K \geq 500$ when an adjusted p lies within 0.03 of α. Across 40 mechanism and size combinations of the scaling runs, the null standard deviation of R_m falls only slowly with mechanism size (log–log slope −0.14), so the same K serves small and large mechanisms.

S6 Exploratory power approximation

For the R and D tests of the primary audit we approximate power at the first Holm step, the most demanding one. Assume the null statistic is normal with the observed null mean $\bar{T}_0$ and s.d. s_0, so that the critical value is $c = \bar{T}_0 + z_{1-\alpha/4}s_0$, and that the observed statistic is normal around its value T with the bootstrap s.d. s_b (conditional on the fitted models). Power is then $\Phi((T-c)/s_b)$, and the minimum excess over $\bar{T}_0$ detectable with 80% power is $z_{1-\alpha/4}s_0 + z_{0.8}s_b$. The approximation gives power near 1 for HDF, PWF, and OSF and 0.00 (R) and 0.02 (D) for TWF; the minimum detectable excess is 0.10–0.14. These numbers rest on normality and on a bootstrap that ignores refitting variability, and are exploratory.

S7 Alternative statistics

Rank-based alignment Rk_m is the per-observation probability that an expected feature outranks a non-expected one in $|\phi|$, averaged over S_m, and Dk_m its difference to other failures (stage 20, $K = 200$). On the primary audit they align PWF and OSF, but HDF fails Rk ($p = 0.33$). Under both misspecified expected sets of the paper, they align no mechanism. They are therefore not less brittle to an incomplete E_m and are less powerful than R_m and D_m.

Outside-set alternatives (i) A studentized maximum (each feature's excess standardized by its own null mean and s.d.) is less sensitive than C_m in the shortcut sweep: it first flags the sensor at $\sigma = 0.20$ ($p = 0.032$), versus $\sigma = 0.25$ for C_m. (ii) A null-referenced score (each outside feature's share on S_m standardized against the null, without other failures) flags TWF in every setting, including the primary audit (on speed or torque, which the model uses for all failures), and misses the sensor at $\sigma = 0.25$ ($p = 0.39$). (iii) C_m computed on the failure-pushing part of SHAP, $\max(\phi, 0)$, does not flag the sensor at any tested strength ($p = 0.139$ at $\sigma = 0.25$), although it flags the omitted triggers.

S8 Several weak proxies

Stage 24 replaces the single planted sensor by three sensors $s_{i,k} = z_{i,\mathrm{HDF}} + \epsilon_{i,k}$ with independent noise of the same σ, all outside E_{HDF} ($K = 500$). Table S4 reports their attribution and the HDF tests. Together the proxies carry more information than one sensor at the same σ, and HDF loses alignment at a larger σ (0.4 versus 0.3). The model spreads attribution over the three proxies, so the largest single excess, which C_m tests, grows slowly; the proxies' summed excess is not tested by C_m. The null mean of C_{HDF} also rises (0.04 to 0.09), because shuffled-label models also attribute to the proxies. Test F1 stays between 0.83 and 0.91, and PWF and OSF remain aligned.

Table S4: Three planted proxies: summed and largest proxy share of HDF attribution, summed proxy excess, C_{HDF} (null mean), and Holm-adjusted p-values for HDF ($K = 500$). Single sensor of the paper for comparison: share 0.13 at $\sigma = 0.35$ (aligned) and 0.35 at $\sigma = 0.25$ (flagged).

σ	Sum share	Max share	Sum excess	C_{HDF} (null)	p_R	p_D / p_C	Verdict
1.0	0.07	0.03	0.00	0.008 (0.040)	.008	.008 / 1.000	aligned
0.67	0.11	0.05	0.03	0.027 (0.058)	.008	.008 / 1.000	aligned
0.5	0.17	0.07	0.09	0.047 (0.073)	.008	.008 / 1.000	aligned
0.4	0.32	0.13	0.25	0.108 (0.082)	.479	.008 / .974	not aligned
0.35	0.38	0.14	0.30	0.114 (0.086)	.838	.008 / .902	not aligned
0.25	0.54	0.25	0.46	0.227 (0.088)	1.000	.182 / .024	flagged

S9 Stability across seeds and backgrounds

Over the primary seed and seeds 1–4, R_m ranges over 0.19–0.20 (TWF), 0.70–0.72 (HDF), 0.68–0.70 (PWF), and 0.66–0.68 (OSF), D_m over at most 0.04, and the three most attributed features of each mechanism are identical in all five runs (pairwise top-3 Jaccard 1.0). Over seven interventional TreeSHAP configurations (100, 500, and 1,000 background observations; three background seeds), R_m varies by at most 0.015 and all verdicts agree. Without the engineered features, the excess of R_m over its null mean falls from 0.04 to 0.02 (TWF), 0.15 to 0.06 (HDF), 0.35 to 0.25 (PWF), and 0.33 to 0.23 (OSF).

S10 Sensitivity analyses

Conditional group Shapley Table S5 varies the kernel bandwidth and the grouping (five groups with tool wear separated from the overstrain features). TWF is aligned in every setting; the other mechanisms lose alignment at bandwidth 0.6, where the kernel averages over most of the data, and HDF's relevant share also under the split.

CWRU defect-frequency bands Table S6 varies the half-width of the defect-frequency bands and adds the third harmonic (original recordings, $K = 1{,}000$).

Table S5: Conditional group Shapley: R_m (null mean) and Holm-adjusted p_R/p_D ($K = 200$); no outside-set test is significant.

Setting	TWF	HDF	PWF	OSF
bandwidth 0.15	0.59 (0.33) .020/.020	0.73 (0.57) .020/.020	0.68 (0.37) .020/.020	0.85 (0.60) .020/.020
bandwidth 0.3	0.54 (0.30) .020/.020	0.63 (0.55) .020/.040	0.65 (0.42) .020/.020	0.80 (0.61) .020/.040
bandwidth 0.6	0.42 (0.28) .020/.020	0.55 (0.51) .129/.239	0.47 (0.38) .104/.060	0.65 (0.57) .129/.239
five groups, 0.3	0.34 (0.21) .020/.020	0.54 (0.48) .070/.030	0.61 (0.40) .020/.020	0.86 (0.66) .020/.030

Table S6: CWRU: R_m (null mean) and Holm-adjusted p_R/p_D for each band definition; no outside-set test is significant.

Bands	Inner race	Ball	Outer race
±0.05 orders, harmonics 1–2	0.45 (0.26) .012/.090	0.08 (0.11) .793/.098	0.42 (0.17) .003/.021
±0.15 orders, harmonics 1–2 (paper)	0.44 (0.23) .004/.030	0.13 (0.17) .773/.211	0.39 (0.15) .003/.027
±0.3 orders, harmonics 1–2	0.41 (0.23) .010/.018	0.22 (0.12) .015/.018	0.36 (0.17) .003/.015
±0.15 orders, harmonics 1–3	0.44 (0.26) .004/.009	0.14 (0.22) .963/.326	0.43 (0.17) .003/.009

Minimum outside-set excess Across all stored results, the significant outside-set flags with excess at most 0.047 are the seven TWF flags on temperatures in the scaling runs, an in-sample TWF flag (0.034), and the TEP IDV5 flag (0.026); every planted sensor, the CWRU gain shortcut, and every omitted trigger has an excess of at least 0.165, and the remaining flags lie between (SmoothGrad on the MLP, 0.074; failure-only background, 0.113). A minimum excess of 0.05 separates the two groups in these data; it was chosen after seeing them and is exploratory (stage 19).

S11 TEP Temporal Models

Adding each variable's 30-minute mean and change as inputs (attribution summed per variable) raises pooled fault-active-class F1 across both fault types to 0.94 and IDV5 recall from 0.39 to 0.86. IDV4 stays aligned (R = 0.68, null 0.07); IDV5's loop share rises to 0.54 (null 0.06), and the E-feed excess shrinks from 0.026 to 0.011 but remains significant, so IDV5 stays flagged. An LSTM (32 units, 20 epochs) on the same samples as 30-minute sequences, explained by Integrated Gradients (64 steps from the training-mean sequence) with attribution summed over time per variable, reaches pooled fault-active-class F1 0.91 across both fault types and IDV5 recall 0.81 (stage 22, K = 200). All TEP variants use the same 400 runs and sampled endpoints, with five stratified group folds holding out complete runs (seed 42); no sequence from a held-out run enters training. For the LSTM, each variable is standardized using the training fold's mean and standard deviation over sequence time steps. The one-layer LSTM is trained with Adam (learning rate 0.003, batch size 256), binary cross-entropy, and a fixed probability threshold of 0.5. Integrated Gradients explains the logit; signed attributions are summed over the ten time steps for each variable before taking the absolute value. Each null draw permutes labels within each run and refits all five models, recomputes training-fold scaling, and re-explains the held-out observations. Mechanism groups retain their original fault-active definitions. This reference does not preserve temporal dependence in the labels and is not a temporal exchangeability guarantee. IDV4 is aligned (R = 0.43, null 0.08), and IDV5 passes both alignment tests (R = 0.23, null 0.06) but is flagged on the D-feed valve (excess 0.011, p = 0.010).

Rotated-label null Within each run the 32 samples are in time order, with 16 normal samples followed by 16 fault-active ones. The rotated-label null shifts each run's label sequence circularly by an offset drawn uniformly from 0–31, so the fault-active labels remain one contiguous block (wrapping around the run's end), and refits and re-explains as above (K = 200; `15_tep_audit.py --null shift`). Table S7 compares it with the within-run permutation. Null means rise slightly, and every verdict is unchanged; all adjusted p-values remain at the minimum or 1. The LSTM was not rerun under this null.

Table S7: TEP under the within-run permutation and the rotated-label null: R_m (null mean) and Holm-adjusted $p_R/p_D/p_C$ (K = 200).

Model	Null	IDV4	IDV5
Static	permutation	0.75 (0.07) .010/.010/1.0	0.22 (0.06) .010/.010/.010
Static	rotation	0.75 (0.08) .010/.010/1.0	0.22 (0.06) .010/.010/.010
30-min features	permutation	0.68 (0.07) .010/.010/1.0	0.54 (0.06) .010/.010/.010
30-min features	rotation	0.68 (0.08) .010/.010/1.0	0.54 (0.07) .010/.010/.010